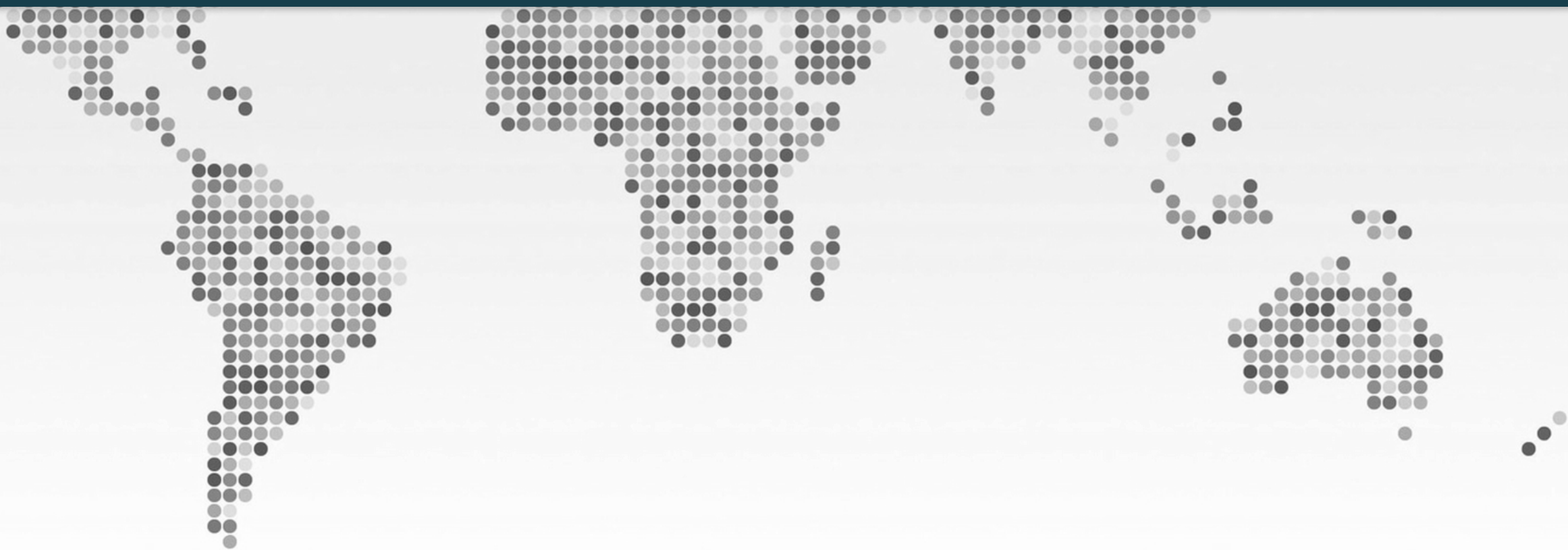


Software Engineering for Artificial Intelligence

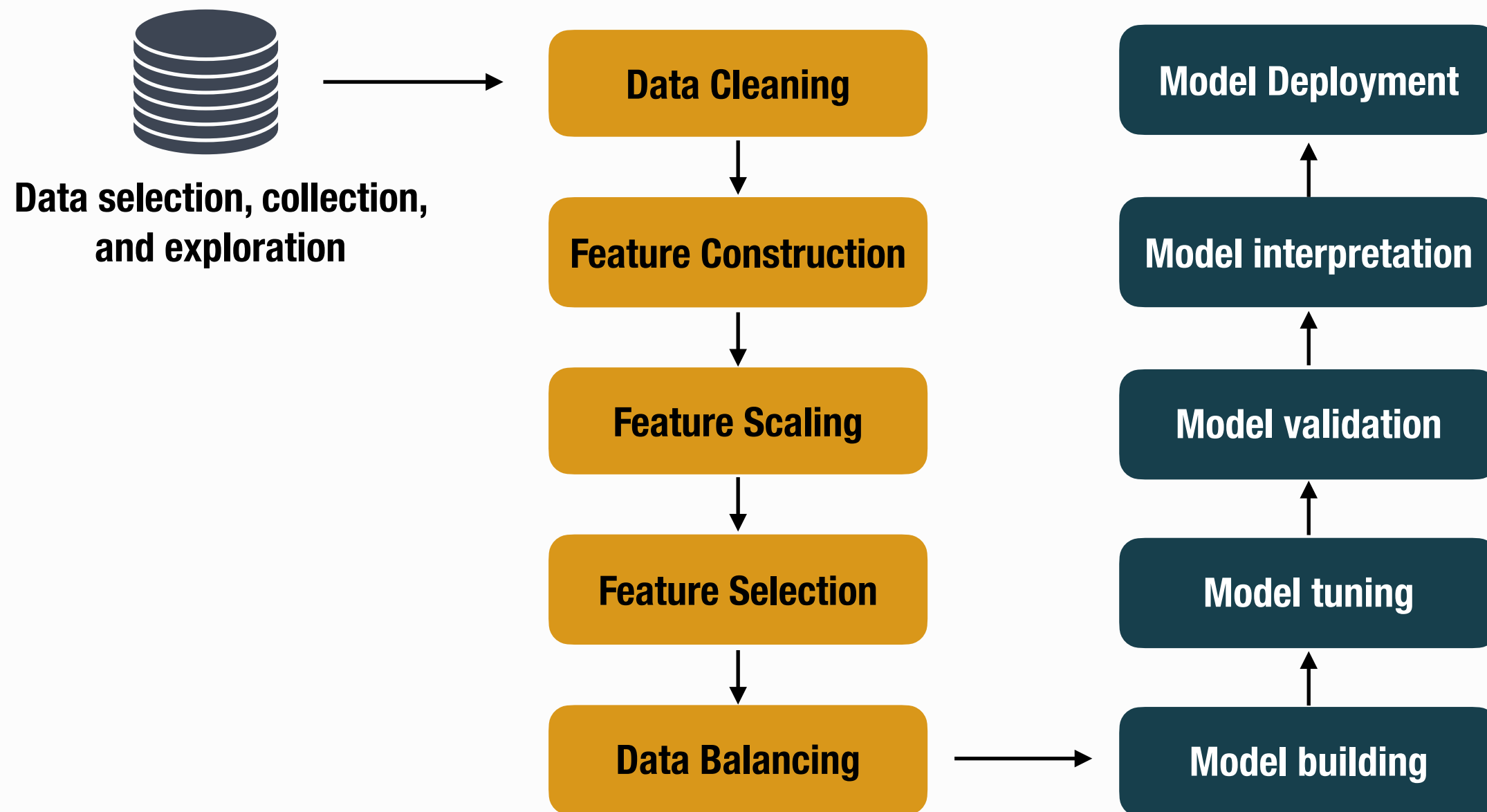
AI as Socio-Technical Systems: Why Models are not Enough



Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

AI as Socio-Technical Systems: Why Models are not Enough

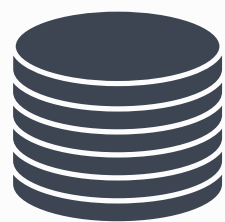
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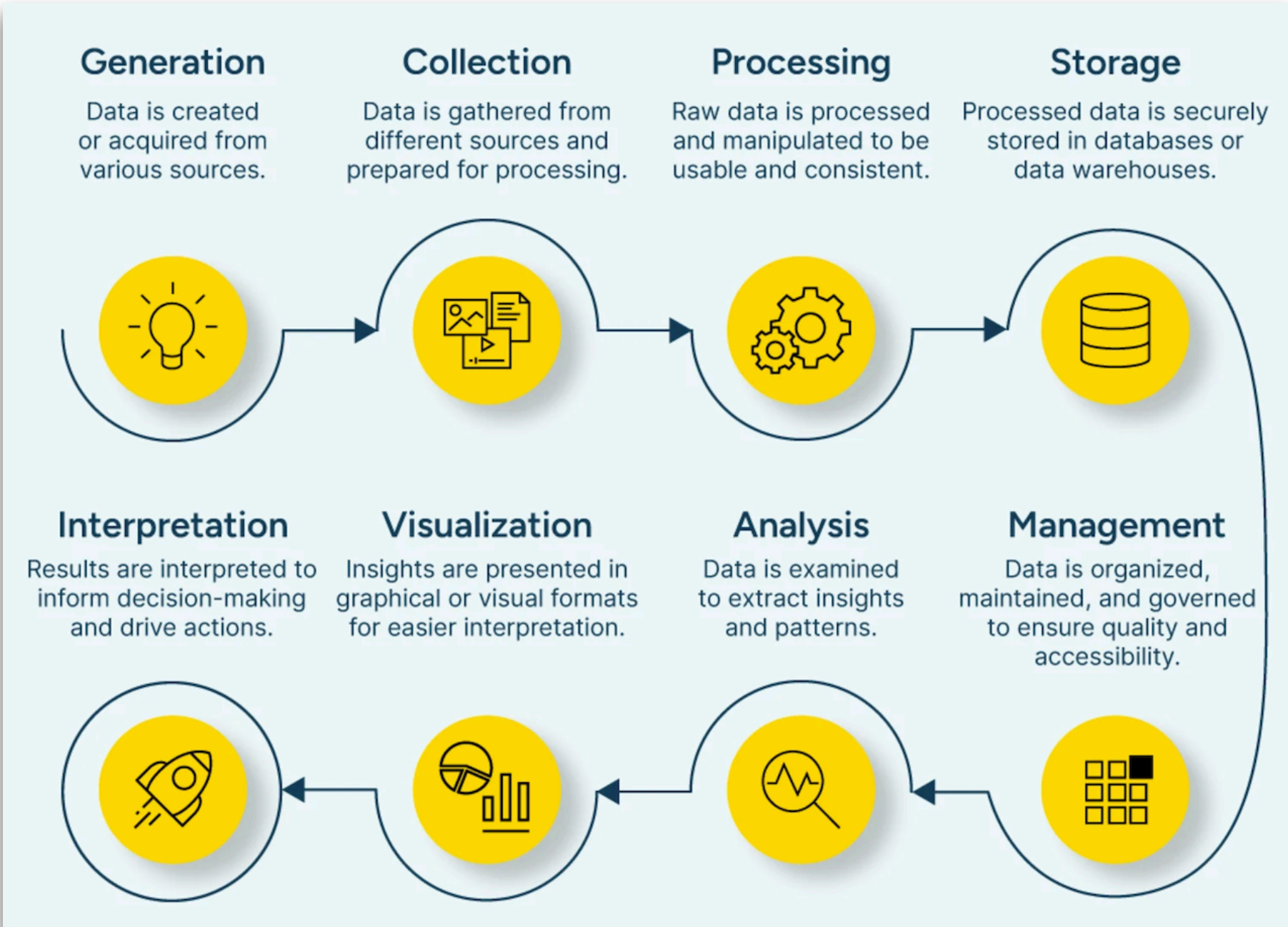
Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

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Data selection, collection, and exploration



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Data selection, collection, and exploration

Data Governance. The management process that curates the availability, usability, integrity, and safety of the data.

Nowadays, data represent one of the major values of companies, organizations, and governments. Implementing strategies to collect, curate, and store data is therefore a urgent need that pertain to the definition of internal and global standards.

The main focus of data governance is concerned with data quality —> good data have higher value and can be used to advance the state of the practice by means of AI!

Data Quality. The set of processes that describe the accuracy, completeness, and consistency of the data collected.

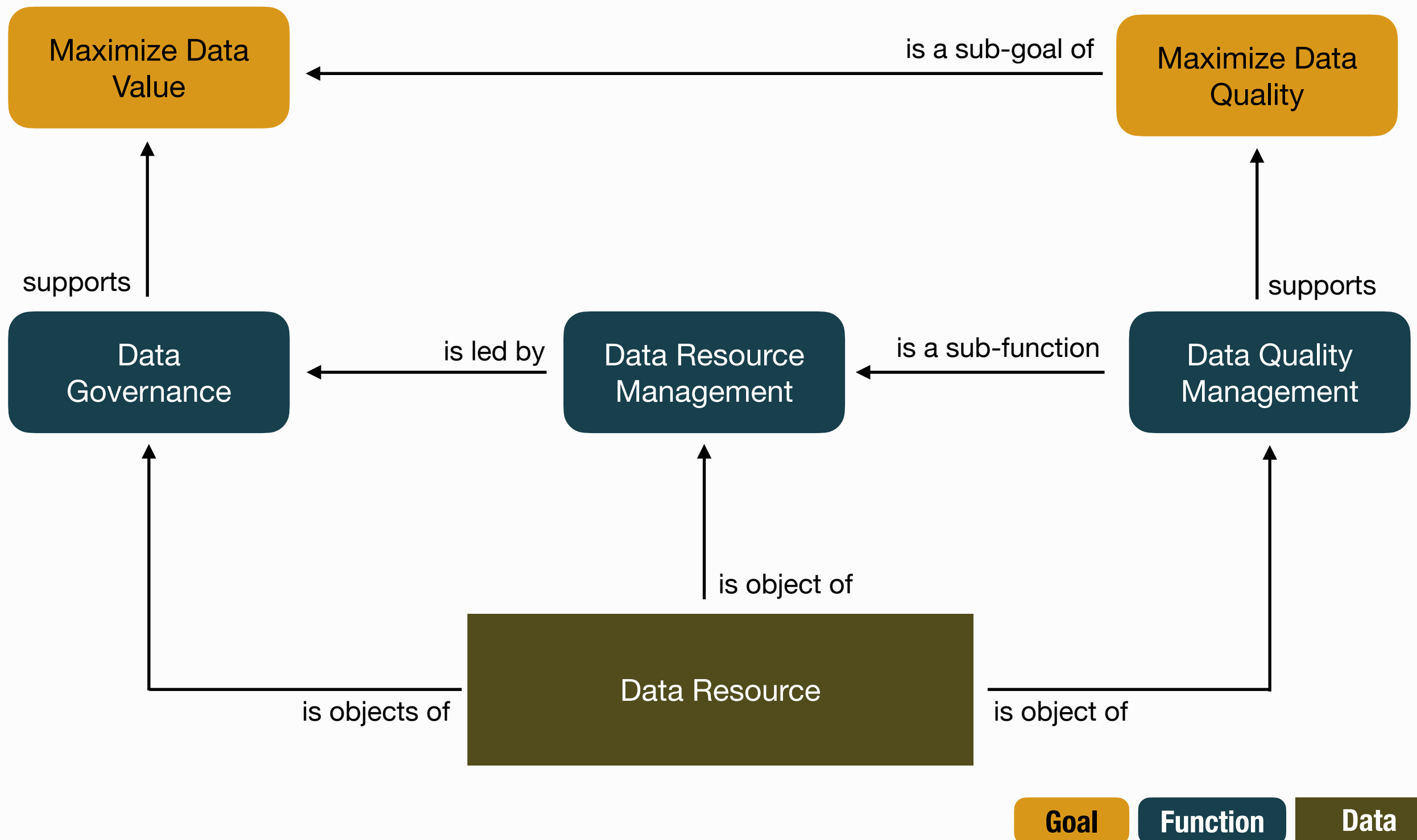
To enable data quality, we refer to data engineering.

Data Engineering. The set of techniques and algorithms that enable the extraction, analysis, and preparation of data. It is the backbone of artificial intelligence.

Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

AI as Socio-Technical Systems: Why Models are not Enough

Let's take a closer look at why data governance is strongly connected to data quality.



Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

AI as Socio-Technical Systems: Why Models are not Enough

A few things to know about data governance...

Data Quality Is Not Accidental

Data governance ensures completeness, consistency, accuracy, timeliness, and traceability. **Model quality is bounded by data quality.** No governance leads to data drifts, inconsistent labeling, and lack of documentation.

Reproducibility & Versioning

Data changes over time, pipelines evolve, labels get updated. Governance supports dataset versioning, control, and tracking. **If you cannot reconstruct the dataset used for training, you cannot audit the model.**

Legal and Regulatory Compliance

Data are bounded by privacy constraints, data protection rules, or sector-specific regulation. **Data governance determines what data you are allowed to use, not just what data you have.**

Bias and Representativeness

Governance affects which populations are included, which variables are collected, how labels are defined, how edge cases are treated. **Bias often originates in governance decisions, not in model architecture.**

Organizational Accountability

Data governance clarifies who owns the dataset, who approves its use, who monitors data quality, who responds to incidents. **In AI systems, data is not just a technical asset, it is an organizational responsibility.**

NB: Even before training, **human and institutional decisions shape system behavior.**

Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

AI as Socio-Technical Systems: Why Models are not Enough

Data: do you really have control over what you can use? **Data governance roles and data engineering roles operate at different layers, but they must align.**

Layer #1: Governance Roles (Decision & Accountability)

The **data owner** is accountable for a dataset, approves usage and access, and is responsible for compliance. Typically, this is a senior domain stakeholder, a business unit manager, a product owner, or department head responsible for the data-generating process. In other terms, **the data owner does not necessarily have engineering and modeling responsibilities.**

The **data steward** ensures data quality, maintains documentation and metadata, and monitors consistency and integrity. Typically, this is a senior data analyst, a data governance specialist, a domain-aware data professional, a quality assurance role within the data team. Unlike the data owner, **the steward is operational and hands-on**: be careful, this is NOT a data scientist!

Layer #2: Engineering Roles (Design & Implementation)

The **data architect** operates at the technical design level, designing storage models, data flows, and integration strategies. **The architect does not define the rules, but must encode them into the system.**

The **database engineer** ensures structural integrity of stored data and implements schema evolution. **Database design operationalizes data quality and compliance decisions.**

The **pipeline engineer** implements preprocessing policies, encodes transformation rules, and applies **validation checks**. **The pipeline engineer operationalizes assumptions about what is 'valid' data.**

Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

AI as Socio-Technical Systems: Why Models are not Enough

Data: do you really have control over what you can use? **Data governance roles and data engineering roles operate at different layers, but they must align.**

Layer #1: Governance Roles (Decision & Accountability)

Layer #2: Engineering Roles (Design & Implementation)

AI models sit at the end of a chain of delegated responsibilities. More importantly, what you learned so far cannot be applied so easily in practice.

In a real-world scenario, you do not simply download a dataset from Kaggle and train a model. Modeling is constrained by a set of governance decisions, regulatory requirements, organizational policies, data availability limitations, and quality constraints.

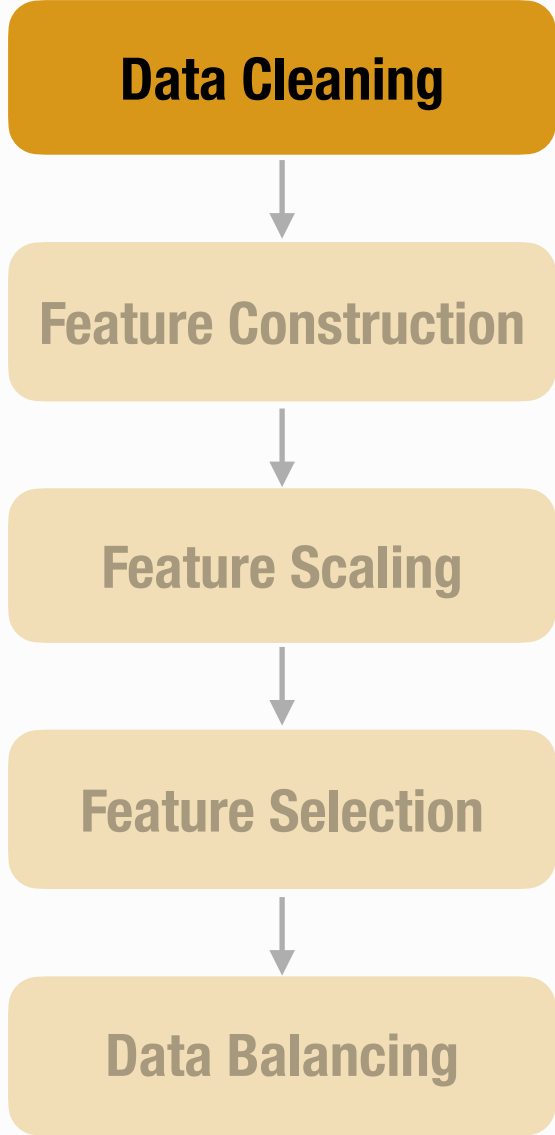
As a model developer, you operate within these constraints and must optimize under them. In other words, **you are rarely solving an unconstrained optimization problem**, you are solving a **socio-technical one**.

Your task is not to build the best theoretical model, but the best model that is legally, organizationally, and technically admissible. This fundamentally changes how we must think about AI beyond simplified settings presented in foundational courses.

Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

AI as Socio-Technical Systems: Why Models are not Enough

Where we left off: Most of you have experienced machine learning as a composition of various elements, including training models on datasets, tuning hyper-parameters, evaluating performance on test data, and comparing models using metrics.



Data Cleaning

When studying data cleaning processes, you have been exposed to typical technical activities, such as data imputation, outliers removal, and similar.

Data cleaning is the **process of deciding what counts as acceptable data**.

This is indeed a mainly technical activity, where you basically deal with noise in the data. Yet, some engineering considerations come into play.

Who defines what counts as an outlier?

Outliers are often defined statistically, e.g., through standard deviations. But think about that: **rare medical cases are outliers, minority groups may appear as statistical anomalies, fraud cases are, by definition, rare.**

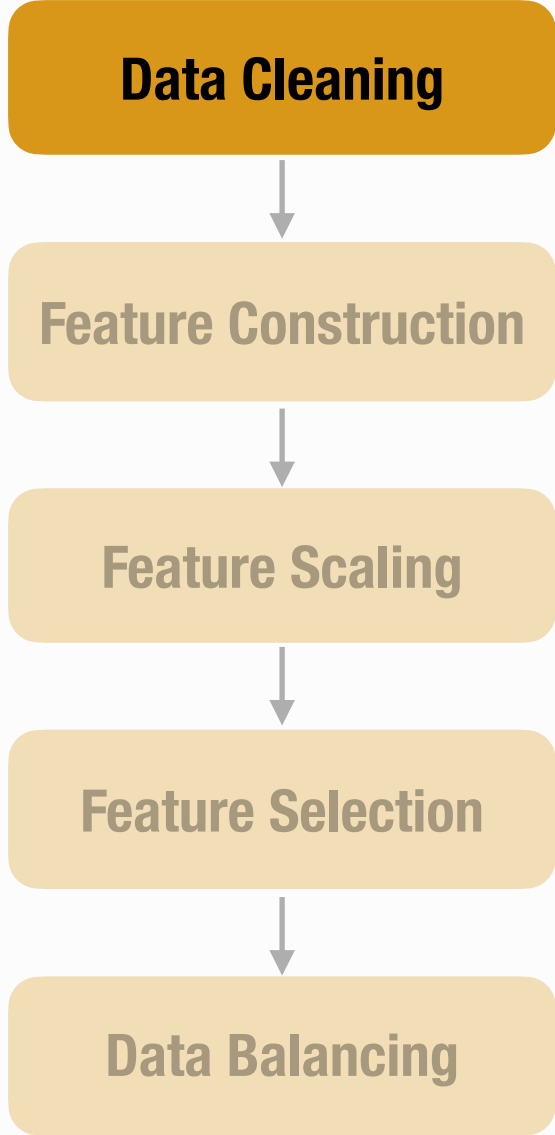
In other terms: in many domains, **the most important cases are precisely the rare ones**. Removing outliers may improve metrics, but possibly degrade real-world value.

Statistical rarity does not imply social irrelevance.
Rare $\neq$ irrelevant, Rare $\neq$ erroneous

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Missing values: error or exclusion?

Missing data might result from measurement failures, incomplete forms, data collection system errors - so, from something technical. But this may also be due to groups less likely to access services, or digital divide, or others.

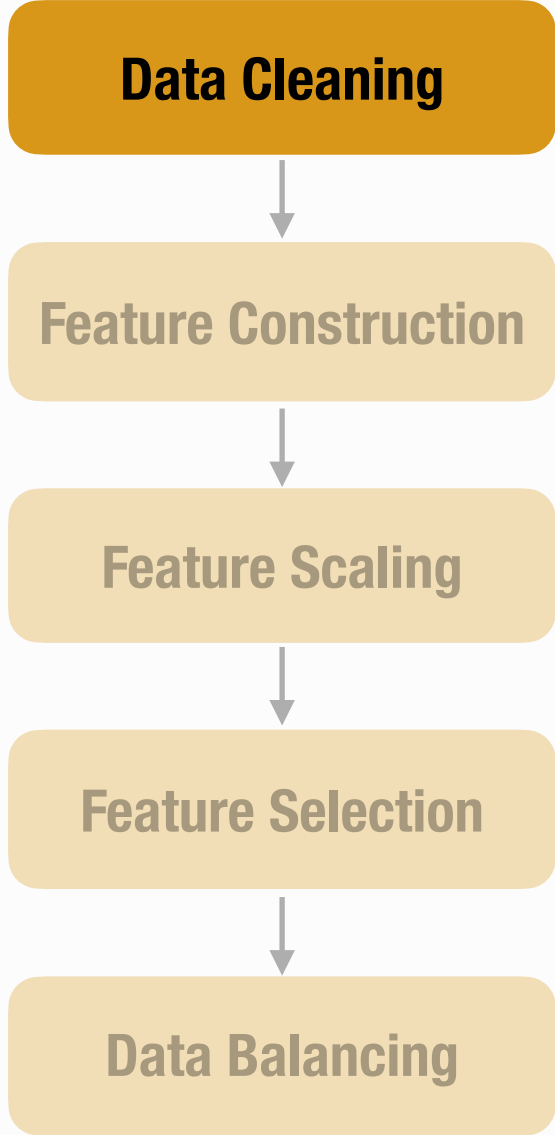
Sometimes missing data is **not noise**: it is evidence of structural inequality.

As a consequence, we cannot blindly operate on missing values: cleaning may become a **political act**, not just a technical one.

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Dropping rows: who disappears?

When we remove records with missing features or inconsistent entries, we may systematically exclude vulnerable populations, edge cases, or complex situations, affecting model generalizability, fairness, and safety!

Data cleaning changes **the population the model is trained on**.

Again, cleaning data may become a **political concern!**

Feature Construction

Feature Scaling

Feature Selection

Data Balancing

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graph TD; A[Data Cleaning] --> B[Feature Construction]; B --> C[Feature Scaling]; C --> D[Feature Selection]; D --> E[Data Balancing];
```

Data Cleaning

Feature Construction

Feature Scaling

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Data Balancing

The Myth of Clean Datasets

The “clean” dataset is not neutral. It reflects human judgments about validity, relevance, and normality.

In other terms, governance shapes what data is collected, cleaning shapes what data survives.

Together, they already construct the system’s **epistemology** - an AI system’s epistemology is the set of assumptions, filters, and representations through which it “knows” reality.

Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

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Data Cleaning



Feature Construction



Feature Scaling



Feature Selection



Data Balancing

From a technical perspective, feature construction involves creating derived variables, aggregating raw measurements, encoding categorical information, or defining composite indicators. The overall goal is to improve model performance and expressiveness.

But also in this case, let's ask ourselves some questions.

What assumptions are embedded in derived features?

What is lost when we aggregate or discretize?

How do we decide what aspects of reality are measurable?

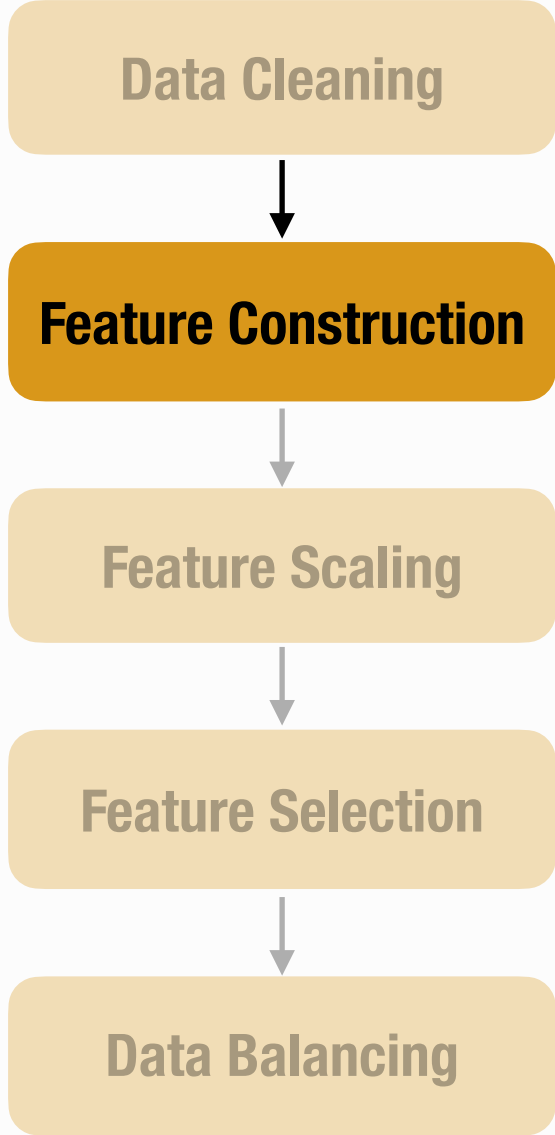
Representation is a **choice**: when you construct features, you decide what to measure, how to measure it, and at what granularity.

Abstraction Hides and Highlights: feature construction always highlights some aspects and hides others, e.g., discretization imposes artificial boundaries.

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From a technical perspective, feature construction involves creating derived variables, aggregating raw measurements, encoding categorical information, or defining composite indicators. The overall goal is to improve model performance and expressiveness.

In other terms, feature construction **determines how reality is compressed into numbers**; when we abstract, we **decide what matters and what does not**.

This power has tremendous responsibilities: before learning patterns, the model learns a representation of reality. **That representation is designed by humans.**

The Myth of Objective Measurement of Reality

Governance shapes what data is collected, cleaning shapes what data survives, feature construction shapes how data is understood.

Feature construction is **epistemic engineering**.

Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

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Data Cleaning

From a technical perspective, feature scaling involves normalization, e.g., min-max, and standardization, e.g., z-score, with the aim of ensuring numerical stability and comparable magnitudes.

Feature Construction

What does it mean to make variables “comparable”?

Are we implicitly equalizing importance across features?

Who decides which variables require scaling?

Feature Scaling

Feature scaling affects **how the model perceives importance**, even if it appears mathematically neutral.

Feature Selection

Standardization as Influence Engineering

Governance shapes what data is collected, cleaning shapes what data survives, feature construction shapes how data is understood, feature scaling shapes how data influences decisions.

Data Balancing

Feature scaling is **geometric engineering**.

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Data Cleaning

Feature selection aims to reduce dimensionality, improve generalization, remove redundancy, and increase efficiency.

Who defines what is “relevant”?

What signals are discarded, and why?

Are we optimizing for accuracy, interpretability, cost, or compliance?

Feature Construction

Feature engineering is not just dimensionality reduction. It is a **decision about what the system is allowed to consider.**

Feature Scaling

Exclusion as Engineering

Governance shapes what data is collected, cleaning shapes what data survives, feature construction shapes how data is understood, feature scaling shapes how data influences decisions, feature selection shapes relevance.

The model does not decide what matters. **We do.**

Feature Selection

Data Balancing

Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

AI as Socio-Technical Systems: Why Models are not Enough

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Data Cleaning

Data balancing aims to address class imbalance, improve performance on minority classes, and stabilize training.

Are we correcting statistical imbalance or redefining reality?

Should the model reflect real-world distributions or ideal ones?

What notion of fairness is being operationalized?

Feature Construction

Feature Scaling

Data balancing is not just statistical adjustment. It is a **decision about how populations are represented in learning.**

Feature Selection

Population as Engineering

Governance shapes what data is collected, cleaning shapes what data survives, feature construction shapes how data is understood, feature scaling shapes how data influences decisions, feature selection shapes relevance, balancing shapes population representation.

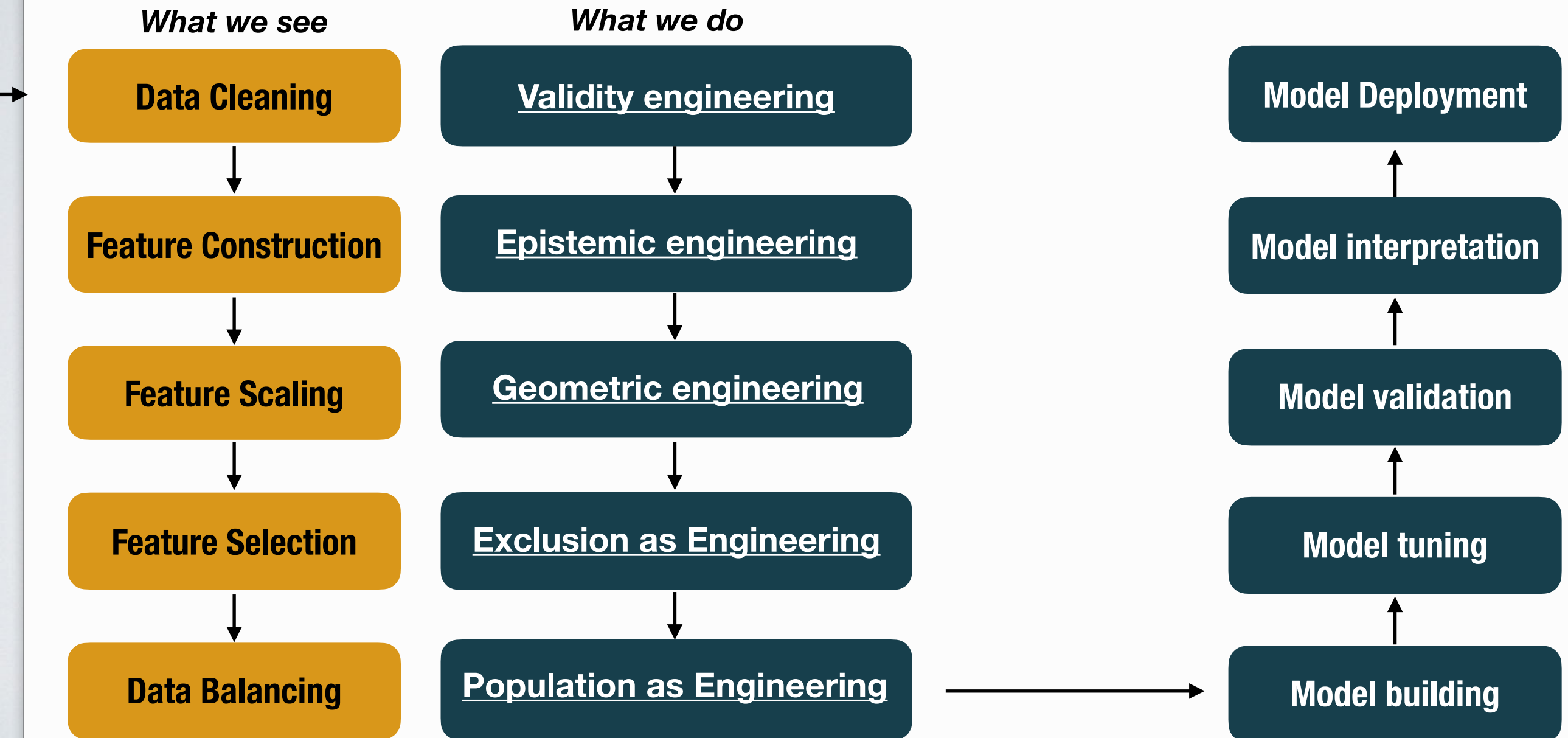
Data Balancing

Data balancing is **distribution engineering.**

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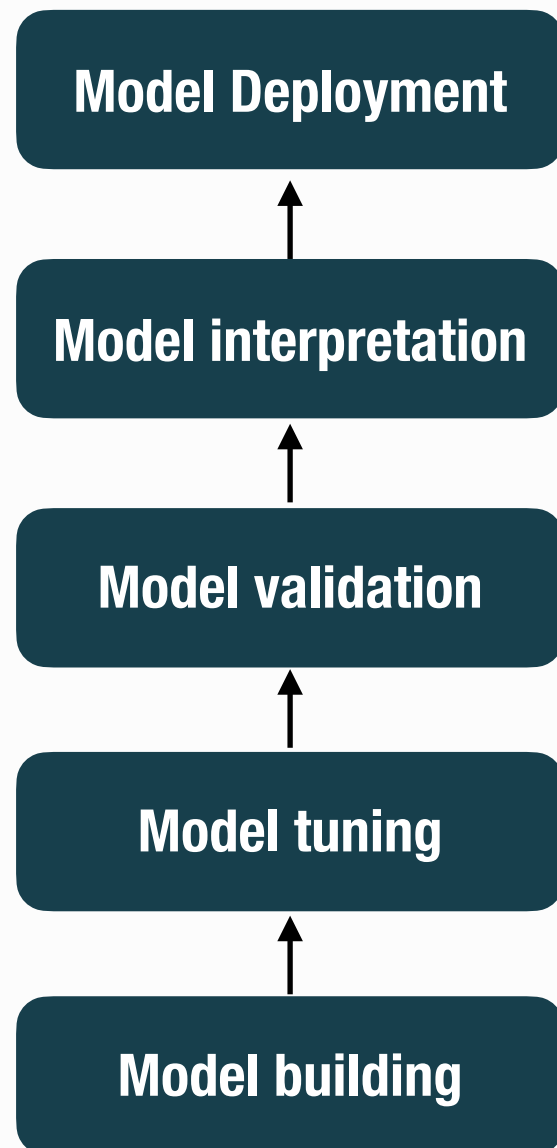


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And even after engineering the dataset, we must choose **objectives**, **architectures**, and **trade-offs**.



Objective Engineering

Objective functions are normative: **the model optimizes what we define as success**, not what is objectively best.

For instance, are we may favor false positives vs false negatives, performance vs interpretability, or performance vs fairness.

Architecture choices reflect assumptions about the nature of the problem: we may choose between peep model vs simple model, black-box vs interpretable model, embedding assumptions about complexity, risk tolerance, deployment constraints, or even organizational expectations.

There is no neutral optimum, **only context-dependent optima**.
Trade-offs are inevitable.

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Engineering

What we see

Model Deployment

Model interpretation

Model validation

Model tuning

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What we do

Objective Engineering

In conclusion, before deployment, we have already engineered **responsibility, validity, representation, geometry, relevance, distribution, and objectives.**

All this just by preparing the data and building the model!

The model is not the beginning of intelligence. It is the culmination of layered socio-technical decisions. In other terms, the artifact is already socio-technical in construction.

Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

AI as Socio-Technical Systems: Why Models are not Enough

Be careful: We have not yet deployed the model.

Deployment changes everything. The moment the model is embedded into an organization, a workflow, or a social context, **it stops being just a predictive artifact.**

It becomes part of a **network** of human decisions, institutional rules, incentives, and feedback loops.

Engineering representations

Engineering interactions

Before deployment

After deployment

Remember, *AI systems are never autonomous*. Even automated systems involve data generators, decision-makers, supervisors, auditors, and affected individuals.

As such, intelligence is distributed across humans and artifacts.

Human-in-the-loop. It describes AI systems in which decision-making authority, responsibility, or oversight is *partially* delegated to algorithms while humans remain structurally involved in shaping, validating, or correcting system behavior.

Human-in-the-Loop does not mean humans are “in control”. Rather, it means humans are *part* of the decision structure.

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Human involvement can occur at different levels.

Humans as Data Generators

Humans generate data in two ways. *Explicitly*, by labeling, annotating, or curating datasets. *Implicitly*, by producing behavioral data, interacting with systems, making or decisions that are recorded.

Humans as Decision Makers

Humans as decision-makers are actors who use AI outputs to inform, support, approve, modify, or execute decisions in real-world contexts.

Humans as Supervisors

Humans as supervisors are actors responsible for monitoring, auditing, correcting, and intervening in AI system behavior over time.

Example: AI-Assisted Radiology System

Role	Actor	What They Shape
Data generators	Radiologists labeling scans	What the model learns
Decision Makers	Radiologist reviewing AI output	What happens to the patient
Supervisors	Hospital AI committee	Whether the system remains trustworthy

Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

AI as Socio-Technical Systems: Why Models are not Enough

The example shows that the AI system is not replacing radiologists: it is **redistributing** expertise, authority, and responsibility across roles.

Indeed, humans delegate part of their *decision authority* to algorithms. Algorithms act within *predefined objectives and constraints*. Humans retain *formal responsibility*, but not full control. In this sense, it is no longer a single actor making a decision, but a **layered process**.

Who
decides?

Power

Delegated Authority. It refers to the process by which humans or organizations assign part of their decision-making power to an algorithmic system, allowing it to influence or determine outcomes under predefined constraints.

Who
acts?

Action

Agency by Proxy. It refers to situations in which AI systems function as operational agents for humans or institutions, producing real-world effects without possessing independent moral or legal agency.

Who
answers?

Accountability

Responsibility Gap. It arises when authority, action, and accountability are distributed across human and algorithmic actors in such a way that no single actor can fully control or explain the resulting outcome.

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Delegation is intentional, but bounded (by objectives, rules, constraints). In any case, delegation does not eliminate human responsibility.

The system acts, but it is not morally autonomous. So, AI systems do not have agency in a *philosophical* sense, but *operational* agency.

Responsibility gaps *are not accidental failures*, but instead structural properties of distributed decision systems due to responsibility fragmentation.

Who
decides?

Power

Delegated Authority. It refers to the assignment of part of their decision-making authority to influence or determine outcomes.

Who
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Action

Agency by Proxy. It refers to the use of operational agents for human actions without possessing independent moral agency.

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Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

AI as Socio-Technical Systems: Why Models are not Enough

As a consequence, once authority is delegated and agency is distributed, interactions become structured, decision pathways become layered, and accountability depends on system design.

In other terms, *the behavior of the system emerges from interactions, not from the model alone*. Very similarly, AI failures are often interaction failures, not algorithmic failures. **This is why models are not enough.**

This means that interactions become part of the engineering process of AI-enabled systems: they shape how outputs are presented, how humans may interpret them, how overrides are allowed, how feedback is captured, how escalation is structured.

This is very nature of Software Engineering for Artificial Intelligence. The real truth is there cannot be AI without software engineering principles.



We will evolve from models to systems when it comes to deploying AI for real world impact,” says Nadella. These systems will have to take into consideration the societal impact they have on people and the planet, he says. “The choices we make about where we apply our scarce energy, compute, and talent resources will matter. ***This is the socio-technical issue we need to build consensus around.***”

<https://economictimes.indiatimes.com/news/new-updates/microsoft-ceo-satya-nadella-calls-for-a-big-ai-reset-in-2026-says-we-need-to-move-beyond-/articleshow/126331455.cms?from=mdr>

Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

AI as Socio-Technical Systems: Why Models are not Enough

Socio-technical AI is a critical shift: it moves our focus from isolated models to the broader networks of humans, institutions, and interactions in which they operate.

Indeed, when interactions are poorly designed, hybrid systems can produce systematic and emergent failures which are beyond models.

Automation Bias

It refers to over-trusting algorithmic outputs.

Causes: Delegated authority reduces scrutiny. Proxy agency creates perceived objectivity. Interfaces present outputs as authoritative.

Over-Reliance and Deskilling

Repeated dependence on AI systems can reduce human expertise over time, decrease independent judgment, increase rare failures.

Causes: Humans become monitors rather than active decision-makers. Responsibility remains, but competence may decline.

Responsibility Diffusion

When outcomes are negative, actors shift blame, accountability becomes unclear, and no one actor feels fully responsible.

Causes: Authority, action, and control are distributed, with no full visibility.

Feedback Loop Amplification

AI decisions influence behavior, behavior influences future data, future data reinforces model patterns. In other terms, *the system begins to shape the reality it learns from.* **Causes:** See above.

Again, these are not accidental failures. They are *emergent properties* of poorly engineered socio-technical interactions.

AI as Socio-Technical Systems: Why Models are not Enough

~~AI is about training models~~
**AI is about engineering distributed
decision systems**

AI We do not engineer models, **we engineer decision architectures.**
We do not optimize metrics, **we design responsibility structures.** **SE4AI**
We do not deploy artifacts, **we embed systems into institutions.**

Technical choices are never purely technical. As *AI engineers*, you are designing representations, encoding normative trade-offs, structuring authority, shaping accountability, defining interaction patterns, and engineering feedback loops.

The real challenge of AI engineering is not building smarter models: **it is designing responsible socio-technical systems.**

As such, intelligence is not in the model. **It is in the system we design around it.**

Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

Before leaving... What Should You Remember?

Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

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Humans generate data in two ways. *Explicitly*, by labeling, annotating, or curating datasets. *Implicitly*, by producing behavioral data, interacting with systems, making or decisions that are recorded.

Humans as Decision Makers

Humans as decision-makers are actors who use AI outputs to inform, support, approve, modify, or execute decisions in real-world contexts.

Humans as Supervisors

Humans as supervisors are actors responsible for monitoring, auditing, correcting, and intervening in AI system behavior over time.

Example: AI-Assisted Radiology System

Role	Actor	What They Shape
Data generators	Radiologists labeling scans	What the model learns
Decision Makers	Radiologist reviewing AI output	What happens to the patient
Supervisors	Hospital AI committee	Whether the system remains trustworthy

Software Engineering for Artificial Intelligence: AI as Socio-Technical Systems

AI as Socio-Technical Systems: Why Models are not Enough

~~AI is about training models-~~

~~AI is about engineering distributed decision systems~~

AI

We do not engineer models, **we engineer decision architectures.**

We do not optimize metrics, **we design responsibility structures.**

We do not deploy artifacts, **we embed systems into institutions.**

SE4AI

Technical choices are never purely technical. As *AI engineers*, you are designing representations, encoding normative trade-offs, structuring authority, shaping accountability, defining interaction patterns, and engineering feedback loops.

The real challenge of AI engineering is not building smarter models: **it is designing responsible socio-technical systems.**

As such, intelligence is not in the model. **It is in the system we design around it.**



Software Engineering for Artificial Intelligence

AI as Socio-Technical Systems: Why Models are not Enough

